
DISCRETE STRUCTURES I

Propositional Logic

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1 Introduction

The rules of logic give precise meaning to mathematical statements. These rules are used to distinguish between valid and invalid mathematical arguments.

Besides the importance of logic in understanding mathematical reasoning, logic has numerous applications to computer science. These rules are used in the design of computer circuits, the construction of computer programs, the verification of the correctness of programs, and in many other ways. Furthermore, software systems have been developed for constructing some, but not all, types of proofs automatically.

2 Propositional Logic

Definition 1. *A proposition is a declarative sentence (that is, a sentence that declares a fact) that is either true or false, but not both.*

Example 1 : All the following declarative sentences are propositions.

1. Washington, D.C., is the capital of the United States of America.
2. Toronto is the capital of Canada.
3. $1 + 1 = 2$
4. $2 + 2 = 3$

Propositions 1 and 3 are true, whereas 2 and 4 are false.

Example 2 : Consider the following sentences :

1. What time is it ?
2. Read this carefully.
3. $x + 1 = 2$.
4. $x + y = z$.

Sentences 1 and 2 are not propositions because they are not declarative sentences. Sentences 3 and 4 are not propositions because they are neither true nor false. Note that each of sentences 3 and 4 can be turned into a proposition if we assign values to the variables.

We use letters to denote propositional variables (or statement variables), that is, variables that represent propositions, just as letters are used to denote numerical variables.

The conventional letters used for propositional variables are $p, q, r, s, \dots$. The truth value of a proposition is true, denoted by T , if it is a true proposition, and the truth value of a proposition is false, denoted by F , if it is a false proposition.

The area of logic that deals with propositions is called the propositional calculus or propositional logic.

We now turn our attention to methods for producing new propositions from those that we already have. Many mathematical statements are constructed by combining one or more propositions. New propositions, called **compound propositions**, are formed from existing propositions using logical operators.

Definition 2. *Let p be a proposition. The negation of p , denoted by $\neg p$ (also denoted by $\bar{p}$), is the statement : "It is not the case that p ".*

The proposition $\neg p$ is read "not p " The truth value of the negation of p , $\neg p$, is the opposite of the truth value of p .

Examples : Find the negation of the following propositions and express them in simple English.

1. p = "Michael's PC runs Linux" :

Solution : The negation is "It is not the case that Michael's PC runs Linux". This negation can be more simply expressed as "Michael's PC does not run Linux".

$\neg p$ = "Michael's PC does not run Linux".

2. q = "Vandana's smartphone has at least 32 GB of memory"

Solution : "It is not the case that Vandana's smartphone has at least 32 GB of memory" or

simply "Vandana's smartphone does not have at least 32 GB of memory, or more simple

q = "Vandana's smartphone has less than 32 GB of memory".

3. r = "At least 10 inches of rain fell today in Manama"

$\neg r$ = "Less than 10 inches of rain fell today in Manama".

The negation of a proposition can also be considered the result of the operation of the negation operator ($\neg$) on a proposition. The negation operator constructs a new proposition from a single existing proposition.

Table 1 displays the truth table for the negation of a proposition p . This table has a row for each of the two possible truth values of a proposition p . Each row shows the truth value of $\neg p$ corresponding to the truth value of p for this row.

We will now introduce the logical operators that are used to form new propositions from two or more existing propositions. These logical operators are also called connectives

Definition 3. *Let p and q be propositions. The conjunction of p and q , denoted by $p \wedge q$, is the proposition "p and q". The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise*

p	$\neg p$
T	F
F	T

TABLE 1 – Truth table for the negation of a proposition

Table 2 displays the truth table of $p \wedge q$. This table has a row for each of the four possible combinations of truth values of p and q . The four rows correspond to the pairs of truth values TT , TF , FT , and FF , where the first truth value in the pair is the truth value of p and the second truth value is the truth value of q .

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

TABLE 2 – The Truth Table for the Conjunction of Two Propositions.

Note that in logic the word "but" sometimes is used instead of "and" in a conjunction. For example, the statement "The sun is shining, but it is raining" is another way of saying "The sun is shining and it is raining." (In natural language, there is a subtle difference in meaning between "and" and "but"; we will not be concerned with this nuance here.)

Example : Find the conjunction of the propositions p and q where p is the proposition "Today is friday" and q is the proposition "It is raining today".

The conjunction of these propositions, $p \wedge q$, is the proposition "Today is Friday and it is raining today". This proposition is true on rainy Fridays and is false on any day that is not a Friday and on Fridays when it does not rain.

Definition 4. Let p and q be propositions. The disjunction of p and q , denoted by $p \vee q$, is the proposition " p or q ". The conjunction $p \wedge q$ is false when both p and q are false and is true otherwise

Table 3 displays the truth table for $p \vee q$.

Example : Find the disjunction of the propositions p and q where p is the proposition "Today is Friday" and q is the proposition "It is raining today".

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

TABLE 3 – The Truth Table for the Disjunction of Two Propositions.

The disjunction of these propositions, $p \vee q$, is the proposition "Today is Friday or it is raining today". This proposition is true on any day that is either a Fridays or a rainy day (including rainy Friday).It is false only on days that are not Fridays when it is also does not rain.

The use of the connective or in a disjunction corresponds to one of the two ways the word or is used in English, namely, as an inclusive or. A disjunction is true when at least one of the two propositions is true. For instance, the inclusive or is being used in the statement "Students who have taken calculus or computer science can take this class". Here, we mean that students who have taken both calculus and computer science can take the class, as well as the students who have taken only one of the two subjects.

However, when a menu at a restaurant states, "Soup or salad comes with an entrée" the restaurant almost always means that customers can have either soup or salad, but not both. Hence, this is an exclusive, rather than an inclusive, or.

Definition 5. Let p and q be propositions. The exclusive or of p and q , denoted by $p \oplus q$, is the proposition that is true when exactly one of p and q is true and is false otherwise.

The truth table for the exclusive or of two propositions is displayed in Table 4.

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

TABLE 4 – The Truth Table for the Exclusive or of Two Propositions.

Exercises :

Question 1. Which of these sentences are propositions? What are the truth values of those that are propositions?

1. Boston is the capital of Massachusetts.
2. Miami is the capital of Florida.
3. $2 + 3 = 5$.
4. $5 + 7 = 10$.
5. $x + 2 = 11$.
6. Answer this question.
7. Do not pass go.
8. What time is it?
9. There are no black flies in Maine.
10. $4 + x = 5$.
11. The moon is made of green cheese.
12. $2^n \geq 100$.

Question 2. What is the negation of each of these propositions?

1. Mei has an MP3 player.
2. There is no pollution in New Jersey.
3. $2 + 1 = 3$
4. The summer in Maine is hot and sunny.
5. Jennifer and Teja are friends.
6. There are 13 items in a baker's dozen.
7. Abby sent more than 100 text messages every day.
8. 121 is a perfect square.

Conditional Statements

We will discuss several other important ways in which propositions can be combined.

Definition 6. Let p and q be propositions. The conditional statement $p \rightarrow q$ is the proposition "if p , then q ". The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise. In the conditional statement $p \rightarrow q$, p is called the hypothesis (or antecedent or premise) and q is called the conclusion (or consequence).

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

TABLE 5 – The Truth Table for the Conditional Statement.

The statement $p \rightarrow q$ is called a conditional statement because $p \rightarrow q$ asserts that q is true on the condition that p holds. A conditional statement is also called an implication. The truth table for the conditional statement $p \rightarrow q$ is shown in Table 5. Note that the statement $p \rightarrow q$ is true when both p and q are true and when p is false (no matter what truth value q has).

Because conditional statements play such an essential role in mathematical reasoning, a variety of terminology is used to express $p \rightarrow q$. You will encounter most if not all of the following ways to express this conditional statement. These different ways are displayed in the table 6.

"if p , then q "	" p implies q "
"if p , q "	" p only if q "
" p is sufficient for q "	"a sufficient condition for q is p "
" q if p "	" q whenever p "
" q when p "	" q is necessary for p "
"a necessary condition for p is q "	" q follows from p "
" q unless $\neg p$ "	

TABLE 6 – Different Ways to Express $p \rightarrow q$

Example 1 :

A useful way to understand the truth value of a conditional statement is to think of an obligation or a contract. consider a statement that a professor might make :

"If you get 100% on the final, then you will get an A"

If you manage to get a 100% on the final, then you would expect to receive an A. If you do not get 100% you may or may not receive an A depending on the professor. However, if you do get 100%, but the professor does not give you an A, you will feel cheated.

Example 2 :

Let p be the statement "Maria learns discrete mathematics" and q the statement

"Maria will find a good job." Express the statement $p \rightarrow q$ as a statement in English : By definition $p \rightarrow q$ is expressed : "If Maria learns discrete mathematics, then she will find a good job."

But, there are many other ways to express this conditional statement in English. Among the most natural of these are :

"Maria will find a good job when she learns discrete mathematics."

"For Maria to get a good job, it is sufficient for her to learn discrete mathematics."

"Maria will find a good job unless she does not learn discrete mathematics."

Remark :

Note that the way we have defined conditional statements is more general than the meaning attached to such statements in the English language.

Example : the proposition "If today is Friday, then $2 + 3 = 6$ " is true for every day except for Friday.

Converse, Contrapositive and Inverse

We can form some new conditional statements starting with a conditional statement $p \rightarrow q$.

Definition 7. The proposition $q \rightarrow p$ is called the converse of $p \rightarrow q$.

The proposition $\neg q \rightarrow \neg p$ is called the contrapositive of $p \rightarrow q$.

The proposition $\neg p \rightarrow \neg q$ is called the inverse of $p \rightarrow q$.

Example :

Write the contrapositive, the inverse and the converse of the following conditional statement :

"The home team wins whenever it is raining"

Let p ="it is raining" and q ="the home team wins".

The converse : $q \rightarrow p$: if q , then p . : "if the home team wins, then it is raining".

The inverse : $\neg p \rightarrow \neg q$: "if it is not raining, then the home does not win"

The contrapositive : $\neg q \rightarrow \neg p$: "if the home team does not win, then it is not raining".

Definition 8. Let p and q be propositions. The biconditional statement $p \leftrightarrow q$ is the proposition " p if and only if q ". The biconditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise. Biconditional statements are also called bi-implications.

The truth table for $p \leftrightarrow q$ is shown in Table 7. Note that the statement $p \leftrightarrow q$ is true when both the conditional statements $p \rightarrow q$ and $p \rightarrow q$ are true and is false otherwise. There are some other common ways to express $p \leftrightarrow q$:

- " p is necessary and sufficient for q "
- "if p then q , and conversely"
- " p iff q ."

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

TABLE 7 – The Truth Table for the biconditional statement.

Example :

Let p be the statement "You can take the flight", and let q be the statement "You buy a ticket". Then $p \leftrightarrow q$ is the statement : "You can take the flight if and only if you buy a ticket".

Exercises :

Question 1. Write each of these statements in the form "if p , then q " in English :

1. It snows whenever the wind blows from the northeast
2. The apples trees will born bloom if it stays warm for a week
3. That the Pistons win the championship implies that they beat the Lakers
4. If you drive than 400 miles, you will need to buy gasoline
5. Your guarantee is good only if you bought your CD player less than 90 days ago
6. Jan will go swimming unless the water is too cold

Question 2. State the converse, contrapositive, and inverse of each of these conditional statements :

1. If it snows today, I will ski tomorrow
2. I come to class whenever there is going to be quiz
3. A positive integer is a prime only if it has no divisors other than 1 and itself.

3 Truth Table of Compound Propositions

We have now introduced four important logical connectives-conjunctions, disjunctions, conditional statements, and biconditional statements as well as negations. We can use these connectives to build up complicated compound propositions involving any number of propositional variables. We can use truth tables to determine the truth values of these compound propositions. We use a separate column to find the truth value of each compound expression that occurs in the compound proposition as it

is built up. The truth values of the compound proposition for each combination of truth values of the propositional variables in it is found in the final column of the table.

Example :

Construct the truth table of the compound proposition $(p \vee \neg q) \rightarrow (p \wedge q)$.

Because this truth table involves two propositional variables p and q , there are four rows in this truth table, one for each of the pairs of truth values TT , TF , FT , and FF . The first two columns are used for the truth values of p and q , respectively. In the third column we find the truth value of $\neg q$, needed to find the truth value of $p \vee \neg q$, found in the fourth column. The fifth column gives the truth value of $p \wedge q$. Finally, the truth value of $(p \vee \neg q) \rightarrow (p \wedge q)$ is found in the last column. The resulting truth table is shown in Table 8.

p	q	$\neg q$	$p \vee \neg q$	$p \wedge q$	$(p \vee \neg q) \rightarrow (p \wedge q)$
T	T	F	T	T	T
T	F	T	T	F	F
F	T	F	F	F	T
F	F	T	T	F	F

TABLE 8 – The Truth Table of $(p \vee \neg q) \rightarrow (p \wedge q)$.

Precedence of Logical Operators We can construct compound propositions using the negation operator and the logical operators defined so far. We will generally use parentheses to specify the order in which logical operators in a compound proposition are to be applied. For instance, $(p \vee q) \wedge (\neg r)$ is the conjunction of $p \vee q$ and $\neg r$. However, to reduce the number of parentheses, we specify that the negation operator is applied before all other logical operators. This means that $\neg p \wedge q$ is the conjunction of $\neg p$ and q , namely, $(\neg p) \wedge q$, not the negation of the conjunction of p and q , namely $\neg(p \wedge q)$.

Table 9 displays the precedence levels of the logical operators : $\neg$, $\wedge$, $\vee$, $\rightarrow$, and $\leftrightarrow$.

4 Translating English Sentences

There are many reasons to translate English sentences into expressions involving propositional variables and logical connectives. In particular, English (and every other human language) is often ambiguous. Translating sentences into compound statements (and other types of logical expressions, which we will introduce later in this chapter) removes the ambiguity. Once we have translated sentences from English into logical expressions we can analyze these logical expressions to determine their truth values, we can manipulate them, and we can use rules of inference to reason about

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

TABLE 9 – Precedence of Logical Operators.

them.

Given a sentence in English, the process consists to decompose the whole sentence in a set of elementary part such that each part (elementary sentence) declares (contains) only one fact.

Then, represent each part by a propositional variable and choose the right (appropriate) logical operators between them to compose the whole meaning of the sentence.

Examples :

1. Consider the sentences : "it is below freezing and snowing", "it is below freezing but not snowing". Let consider the propositional variables : p ="it is below freezing" and q ="it is raining". So we can write : $p \wedge q$, $p \wedge \neg q$.
2. How can this English sentence be translated into a logical expression? "You can access the Internet from campus only if you are a computer science major or you are not a freshman." we let a , c , and f represent "You can access the Internet from campus", "You are a computer science major", and "You are a freshman" respectively.
 $a \rightarrow c \vee \neg f$.

5 Propositional Equivalences

Definition 9. A compound proposition that is always true, no matter what the truth values of the propositional variables that occur in it, is called a tautology. A compound proposition that is always false is called a contradiction. A compound proposition that is neither a tautology nor a contradiction is called a contingency.

Example :

Definition 10. The compound propositions p and q are called logically equivalent if $p \leftrightarrow q$ is a tautology. The notation $p \equiv q$ denotes that p and q are logically equivalent.

Examples :

Using truth table, show that :

1. $\neg(p \vee q)$ and $(\neg p \wedge \neg q)$ are logically equivalent.

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

TABLE 10 – Example Tautology, Contradiction

2. $p \rightarrow q$ and $\neg p \vee q$ are equivalent.
3. $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are equivalent.
4. $p \rightarrow q$ and $\neg q \rightarrow \neg p$ are equivalent.

Consider the sentence "if it snows today, then i will ski tomorrow". This sentence is equivalent to "it does not ski today or i will ski tomorrow".

Exercise : Give the OR form of the different conditional statements proposed in the exercise 1.

Theorem 1 (DeMorgan Law's).

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

Proof :

Show, using truth table :

1. $\neg(p \wedge q)$ and $\neg p \vee \neg q$ are equivalent.
2. $\neg(p \vee q)$ and $\neg p \wedge \neg q$ are equivalent.

Examples :

Use De Morgan's laws to express the negations of "Miguel has a cellphone and he has a laptop computer" and "Heather will go to the concert or Steve will go to the concert".

Let p be "Miguel has a cellphone" and q be "Miguel has a laptop computer". Then "Miguel has a cellphone and he has a laptop computer" can be represented by $p \wedge q$. By the first of De Morgan's laws, $\neg(p \wedge q)$ is equivalent to $\neg p \vee \neg q$. Consequently, we can express the negation of our original statement as "Miguel does not have a cellphone or he does not have a laptop computer."

Let r be "Heather will go to the concert" and s be "Steve will go to the concert". Then "Heather will go to the concert or Steve will go to the concert" can be represented by $r \vee s$. By the second of De Morgan's laws, $\neg(r \vee s)$ is equivalent to $\neg r \wedge \neg s$. Consequently, we can express the negation of our original statement as "Heather will not go to the concert and Steve will not go to the concert".

6 Constructing New Logical Equivalences

The established logical equivalences, can be used to construct additional logical equivalences. The reason for this is that a proposition in a compound proposition can be replaced by a compound proposition that is logically equivalent to it without changing the truth value of the original compound proposition.

Examples : Without using a truth table,

1. Show that $\neg(p \rightarrow q)$ and $p \wedge \neg q$ are logically equivalent.
2. Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.
3. Show that $\neg(p \vee (\neg p \wedge q)) \equiv (\neg p \wedge \neg q)$
4. Show that $(p \wedge q) \rightarrow p$ is tautology
5. Show that $\neg p \rightarrow (p \rightarrow q)$ is tautology

7 Exercises

Exercise 1 :

Consider the following propositions :

d = : "You drop the cup"

b = : "The cup will break"

t = : "The cup will fall over the blanket"

c = : "The cat will run away with blanket"

Answer the following :

Question 1. Give the statement $\neg b \rightarrow \neg d \wedge \neg c$ in English using *whenever*.

Question 2. Give the inverse in symbolic form for the statement :

A necessary condition for you drop the cup and the cup will not break is the cat will not run a way with blanket and the cup will fall over the blanket.

Question 3. Give the converse in English for the following statement $c \vee \neg t \rightarrow b$

Question 4. Give the symbolic for : "the cup will break unless it will fall over the blanket and the cat will not run away with blanket".

Question 5. Give the English using unless : $d \wedge b \rightarrow \neg t \vee c$

Question 6. Give the negation in symbolic form for the following statement using (and, or, and) only.

the cat run a way with blanket or you do not drop the cup is sufficient for the the cup will not break.

Question 7. Give the negation in English : $\neg b \rightarrow t \wedge \neg c$

Question 8. Give in English using only if : $(\neg d \vee \neg b) \vee \neg t$.

Question 9. Write the symbolic Or form : *you do not drop the cup unless neither the cup will break nor it will fall over the blanket.*

Question 10. Write in English the contrapositive for :
the cat will not run a way with blanket or you will not drop it is necessary for the the cup will not break and it will not fall over the blanket

Exercise 2 :

Let p, q, r, s denote the following statements :

p : Jack ate the apple.

q : Jack ate the candy.

r : Jack is hungry.

s : Jack is wise.

Answer the following

Question 1. Write in English using follows form for : $p \wedge \neg q \longrightarrow s \vee r$.

Question 2. Write the symbolic for : *Jack neither hungry nor wise if he ate neither the apple nor the candy.*

Question 3. Rewrite part (2) using unless.

Question 4. Write the inverse in English using A sufficient condition for : $r \wedge q \longrightarrow \neg s$.

Question 5. Write the converse in symbolic of : *Jack did not eat the candy unless he is either hungry or not wise* .

Question 6. Write the negation in English. Do not use "it is not the case" : *A necessary condition for jack did not eat both the candy and the apple is he is not hungry or not wise* .

Question 7. Write the contrapositive using implies : $p \wedge q \longrightarrow r \wedge \neg s$.

Question 8. Write the or form in English : *Jack ate the candy but did not eat the apple unless he is wise and not hungry* .

Question 9. Give the symbolic for : *Jack ate both the apple and the candy whenever he is not hungry but wise* .

Exercise 3 :

Question 1. Construct the truth table of :

1. $p \oplus (p \vee q)$

2. $(p \leftrightarrow q) \oplus (p \leftrightarrow \neg q)$

3. $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$

Question 2. State each of these conditional statements is a tautology or not

1. $(\neg p \wedge (p \vee q)) \rightarrow q$

2. $(p \wedge q) \rightarrow r \leftrightarrow (p \rightarrow r) \wedge (q \rightarrow r)$

Exercise 4 :

Question 1. The father said : "The baby will cry unless you give him food or play with him". The mother said : your wrong the right statement is "You neither play with baby nor give him food or the baby will cry".

Is the father statement the same as the mother or not ?

Question 2. Ahmed has two sons Adel and Kareem. Ahmed said to his sons "the cat will neither chase nor eat the mouse unless the mouse will squeak". Adel told his mother, my father said "A necessary condition for the cat either chase or eat the mouse is he will not squeak". However Kareem told his mother, my father said "the mouse will squeak is sufficient for the cat to neither eat nor chase the cat".

Find out who told his mother the correct statement the father said.

Exercise 5 :

Question 1. Show that $(p \wedge q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$ are not logically equivalent.

Question 2. Show whether the following are logically equivalent or not.

$p \longrightarrow r \vee q$ and $p \wedge q \longrightarrow r$

Question 3. Prove using logical equivalence (law) that $(p \wedge q) \wedge (p \rightarrow r) \rightarrow (q \vee r)$ is a tautology.

Question 4. Show that $p \longrightarrow q \longleftrightarrow \neg q \longrightarrow \neg p$ is a tautology

Question 5. Show by truth a assignment that $\neg p \wedge \neg q \longleftrightarrow r \oplus \neg s$ is not equivalent to

$[(\neg r \vee s) \wedge (r \vee \neg s) \vee (\neg p \wedge \neg q)] \wedge [p \vee q \longrightarrow (r \vee \neg s) \wedge \neg(r \wedge \neg s)]$